

BAD601 TIE SIMP Questions - based on 22 scheme

Module 1

1. Explain the different classifications of data – structured, unstructured, and semi-structured – with suitable examples and sources.
2. What are the key characteristics of data? Discuss composition, condition, context, and the six C's (accuracy, validity, reliability, timeliness, relevance, completeness).
3. Define Big Data and explain its evolution. Include the 3 V's (Volume, Velocity, Variety) and additional V's like Veracity, Volatility, Variability.
4. Why is Big Data important for modern organizations? List and explain at least five real-world benefits (e.g., customer retention, targeted marketing, operational efficiency).
5. Explain the classification of analytics into basic, operationalized, advanced, and monetized analytics..
6. Compare Traditional Business Intelligence and Big Data. Include differences in architecture, processing approach, scalability, and type of data handled.
7. Describe the key technologies used in Big Data Environments such as:
 - In-memory analytics
 - In-database processing
 - Massively Parallel Processing (MPP)
 - Shared Nothing Architecture
 - CAP Theorem
8. What is Hadoop? Explain its features, architecture (HDFS + MapReduce/YARN), and key ecosystem components (Hive, Pig, HBase, etc.).
9. Explain NoSQL databases. Discuss types (key-value, document, columnar, graph), characteristics, CAP theorem trade-offs, and industry use cases.

Module 3

1. What is MongoDB? Explain its architecture, document model, BSON, and differences from traditional RDBMS.
2. Compare RDBMS and MongoDB.

3. What are the key data types in MongoDB? Describe with examples: String, Number, Array, ObjectId, Date, Null, Binary, RegEx, and more.
4. Explain CRUD operations in MongoDB using shell commands. Include examples for insert(), find(), update(), remove(), save(), and replaceOne().
5. How do you perform filtering, sorting, projection, and pagination in MongoDB?
6. How are arrays managed and queried in MongoDB? Explain with examples using \$size, \$all, \$slice, \$push, \$pop, \$pull, and positional updates.
7. What is the aggregation pipeline in MongoDB? Explain stages like \$match, \$group, \$sum, \$avg, \$max, \$min, and include use cases.
8. Explain MapReduce in MongoDB with a custom example. Show how emit, map, and reduce functions work together.
9. How is indexing handled in MongoDB? Explain ensureIndex(), types of indexes, covered queries, and how to analyze index usage with explain().

Module 4

1. What is Apache Hive? Explain its architecture, features, components (Metastore, Driver, CLI), and how it integrates with Hadoop.
2. Differentiate between internal (managed) and external Hive tables. Explain syntax, storage location, and behavior during DROP.
3. Explain partitioning and bucketing in Hive. Describe static vs dynamic partitioning and when to use each. Include CLUSTERED BY example for bucketing.
4. What are the major Hive file formats? Compare TextFile, SequenceFile, and RCFile. Explain RCFile's advantage in aggregation-heavy queries.
5. Write HiveQL commands for DDL and DML operation for creating databases, inserting data, describing tables, and loading from local/HDFS.
6. What are User Defined Functions (UDFs) in Hive? How do you write a UDF in Java, convert to JAR, register and use it inside HiveQL?
7. What is Apache Pig? Explain Pig Latin, Pig's architecture (Grunt shell, Execution engine), Pig on Hadoop, and Pig Philosophy.
8. Explain any 4 relational operators in Pig Latin with examples.
9. Compare Pig and Hive. Differences in purpose, language, usage, performance, use cases (ETL vs reporting), and developer roles.

Module 5

1. What is Apache Spark? Describe Spark architecture, main components (Spark Core, SQL, Streaming, GraphX, MLlib), and execution model (in-memory + on-disk).
2. Explain Resilient Distributed Datasets (RDDs) in Spark. What are RDDs? Discuss their properties (fault-tolerant, distributed, immutable), operations (transformations, actions), and role in Spark.
3. What is the Spark Software Stack? Explain how Spark SQL, Streaming, R, MLlib, GraphX, and Arrow form a layered architecture. Give brief uses of each component.
4. Explain the use of Spark SQL
5. What is a DataFrame in Spark? Explain how to create DataFrames from Parquet/JSON/Hive, and how to use filter(), select(), groupBy(), and join() operations.
6. What is a User Defined Function (UDF) in Spark SQL? Explain with a PySpark example that increases a product cost by 10%. Show how to define, register, and use a UDF in a query.
7. What are Vectorized UDFs (VUDFs) and Grouped Vectorized UDFs (GVUDFs)? Explain
8. Compare and explain the role of NumPy, SciPy, and Pandas in Spark-based data analysis.
9. How can Spark + Python be used to remove outliers in a dataset? Show PySpark code that computes mean and standard deviation and filters values within a reasonable range (e.g., $\pm 3\sigma$).

Module 2

1. Explain the limitations of traditional RDBMS in the context of big data, and why Hadoop's architecture is more suitable.
2. Compare RDBMS and Hadoop in terms of architecture, processing, storage, and cost wrt structured vs unstructured data support, scalability, and hardware requirements.
3. Explain the core components of Hadoop architecture.
4. Describe HDFS architecture and its working..
5. What is MapReduce? Explain the Map, Shuffle & Sort, and Reduce phases with an example like Word Count.
6. Explain the roles of Mapper, Combiner, Partitioner, and Reducer in MapReduce programming.

7. What is YARN in Hadoop 2.0? Explain its components: ResourceManager, ApplicationMaster, NodeManager, and how it improves over Hadoop 1.x.
8. Explain the steps involved in reading and writing a file in HDFS. Explain anatomy of file read/write and communication between client, NameNode, and DataNode.
9. Write and explain Hadoop file system commands for:
 - Creating directories
 - Uploading/downloading files
 - Viewing content
 - Removing directories
(e.g., *hadoop fs -mkdir, -put, -get, -cat, -rm -r*)